

SECTION 26 05 43
UNDERGROUND DUCTS AND CONTAINMENT FOR ELECTRICAL SYSTEMS (EMEA)

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PART 0 PROJECT-SPECIFIC AMENDMENTS

The following amendments to this Section apply only to this Project:

- 2.1.A: Replace 'Provide duct bank with minimum 150 mm concrete surround wherever ductbank crosses under roadways, car parks and where located in unstable soil and within 1500 mm of building foundations (with suitable separation)' with 'All MV and HV ducts to be in encased in concrete with minimum 150mm surround. LV and ELV ducts to be encased in concrete with minimum 150mm surround where depth of cover is <750mm in paved areas and <500mm in landscaped areas, otherwise minimum 150mm sand surround.'
- 2.1.A: Add 'Number and size of ducts is as per electrical information.'

PART 1 GENERAL

1.1 SUMMARY (NON-INCLUSIVE)

- A. Underground duct bank, manhole and handhole distribution systems for electric power, including required accessories, for a complete and operational system. Provide cables as indicated and make all required connections, unless otherwise indicated within the design package. Manholes and handholes shall be precast or cast-in-place concrete.
- B. This section shall apply to underground single or grouped conduit as indicated on the Drawings as duct sections for feeders.

1.2 RELATED REQUIREMENTS

- A. Applicable Sections: Division 01.
- B. Applicable Sections: Division 03.
- C. Applicable Sections: Division 27.
- D. Applicable Sections: Division 31.
- E. Applicable Sections: Division 33.
- F. Section 26 05 13 - Medium Voltage Cables.
- G. Section 26 05 19 - Low-Voltage Electrical Power Conductors and Cables.
- H. Section 26 05 26 - Earthing and Bonding for Electrical Systems.
- I. Section 26 08 01 - Testing Electrical Systems.
- J. Section 03 30 00 - Cast-in-Place Concrete.
- K. Section 26 05 44 - Sleeves and Sleeve Seals for Electrical Systems.
- L. Section 26 05 33 - Conduit.
- M. Section 26 05 37 - Trunking.

- N. Section 26 05 36 - Cable Trays for Electrical Systems.
- O. Section 26 05 29 - Hangers and Supports for Electrical Systems.
- P. Section 33 71 19 - Underground Conduits, Ducts, Vaults and Manholes.
- Q. Section 26 05 53 - Identification for Electrical Systems.
- R. Section 26 41 13 - Lightning Protection for Structures.
- S. Section 26 05 73 - Power System Studies.
- T. Electrical Design Guide.

1.3 SUBMITTALS

- A. Submit according to the requirements of Division 01.
- B. Shop Drawings, including but not limited to: Show route and details of underground duct bank installation including manhole and handhole details and the duct sweeps with radius called out in this specification and the construction documents.
- C. Product Data: Standard manufacturer's data sheet.
- D. Calculations: Submit Cable-Pulling Plan calculations for all MV power cable and low voltage power cable pulls. Refer to Section 26 05 13 - Medium Voltage Electrical Conductors and Cables and Section 26 05 19 - Low-Voltage Electrical Power Conductors and Cables.
- E. Duct Bank coordination drawings to show duct profiles and coordinated/approved clearances to other underground services/structures as per IEC/EN and local regulations as a minimum.
- F. Soil testing: Contractor shall be responsible for obtaining all soil conditions of final fill (thermal and electrical resistivity) to determine calculations to IEC 60287-2-1 and any local regulations. A report of soil conditions in areas of duct banks shall be provided as a submittal to the A/E of Record prior to install. Soil conditions report to include soil bearing pressure.

1.4 QUALITY ASSURANCE

- A. Material and installation shall comply with all applicable codes and recommended practices and standards of IEEE, IEC, CENELEC, EN, ISO and any additional National Standards that apply in the host EMEA Country.
- B. All components shall be IEC listed, CE marked and labeled:
 - 1. All electrical equipment and systems supplied shall conform to the appropriate regional directives and shall carry the applicable CE marking. The suppliers' documentation shall include declarations of conformity for electrical equipment

- within this scope of supply as required by machinery directive, LV directive, and EMC directive.
2. Compliance of the electrical equipment supplied, with regional standards and the related legislation is solely the responsibility of the supplier regardless of whether the equipment in question is sourced from sub-supplier or in-house. If the supplier deems that this equipment must be further assessed or certified as part of the larger electrical system to which it will finally be connected, then they shall highlight this in their tender for approval from the client, failing which responsibility shall revert to the supplier.
- C. Referenced Codes and Standards: Comply with the following in accordance with Division 01:
1. International Organization for Standardization (ISO)
 - a. EN ISO 9000 - Quality management and quality assurance standards. Guidelines for selection and use.
 - b. EN ISO 9001 - Quality Systems Model for quality assurance in Design, Development, Production, Installation and Servicing.
 - c. EN ISO 1461 - Hot dip galvanized coatings on fabricated iron and steel articles. Specifications and test methods.
 2. International Electrotechnical Commission (IEC), European Norms (EN) and European Directive / Harmonised Documents (HD)
 - a. International Electrotechnical Commission (IEC) / European Norms (EN)
 - 1) IEC 60634 (all parts) - Low Voltage Electrical Installations.
 - 2) IEC 60287-2-1 - Electric cables - Calculation of the current rating - Part 2-1: Thermal resistance - Calculation of the thermal resistance.
 - 3) IEC 60529 - Degrees of Protection Provided by Enclosures (IP Code).
 - 4) IEC 60364-6 - Low voltage electrical installations - Part 6: Verification.
 - 5) IEC 61537 - Cable management - Cable tray systems and cable ladder systems.
 - 6) IEC 62561-2 - Lightning protection system components (LPSC). Requirements for conductors and earth electrodes.
 - 7) IEC 61386-24 - Conduit systems for cable management Part 24: Particular requirements Conduit systems buried underground.
 - 8) IEC 61386 (all parts) - Conduit systems for cable management.
 - 9) EN 50110 - Operation of electrical installations. General requirements.
 - 10) EN 50085 (all parts) - Cable trunking systems and cable ducting systems for electrical installations.
 - 11) EN 13476 - HDPE Conduit.
 - 12) EN 13476 - Specification for Underground Enclosure Integrity.
 - 13) EN 10346 - Continuously hot-dip coated steel flat products for cold forming - technical delivery conditions.
 - 14) EN 1916 - Concrete pipes and fittings, unreinforced, steel fiber and reinforced.
 - 15) EN 681 - Elastomeric seals. Material requirements for pipe joint seals used in water and drainage applications.
 - b. European Directive / Harmonised Documents (HD)
 - 1) 2014/30/EU - Electromagnetic Compatibility Directive.
 - 2) 2014/35/EU - Low Voltage Directive.
 - 3) CENELEC HD 603 S1 - Distribution Cables of Rated Voltage 0.6/1 kV.

- 4) EU Eco Directive.
3. For all reference standards listed in this section, use the latest edition from the date of purchase order unless otherwise specified.
4. The referenced standards, regulations, and directives in this specification shall mean all parts including any addendum and corrigendum. It shall also mean any standards, regulations, and directives referenced within them.
5. International Mechanical Code (IMC) (where appropriate and where in accordance with local codes and ordinances).
6. European Agency for Safety and Health at Work (EU-OSHA).
7. Local Electric Utility Company Rules and Regulations for Underground Electric Service.

PART 2 PRODUCTS

2.1 UNDERGROUND DUCT LINES

- A. Duct lines shall be encased in sand or concrete as per recommendation from cable calculation; see Division 03 Concrete and Section 03 30 00 - Cast-in-Place Concrete for requirements. All MV and HV ducts to be in encased in concrete with minimum 150mm surround. LV and ELV ducts to be encased in concrete with minimum 150mm surround where depth of cover is <750mm in paved areas and <500mm in landscaped areas, otherwise minimum 150mm sand surround. Provide duct bank with minimum 150 mm concrete surround wherever ductbank crosses under roadways, car parks and where located in unstable soil and within 1500 mm of building foundations (with suitable separation), and as otherwise indicated within the design package:
 1. Provide plastic warning tape above the duct bank along the entire route. Duct Banks shall consist of individual high-density polyethylene (HDPE) conduits, 90 degrees C, 450/750N as per IEC 61386-24, not smaller than 100 mm in diameter unless otherwise indicated on the Drawings. Number and size of ducts is as per electrical information. Ducts shall be as manufactured by:
 - a. Selected and approved EU local Manufacturers.
- B. Provide continuous 1000 volt stranded copper earthing conductor in each duct in the bank used for feeder power cables. Earthing conductor shall be installed with the current carrying conductors. The size of the earthing conductor shall be minimum 150 mm² (or to local regulation, whichever is greater) unless otherwise indicated within the design package; refer to Section 26 05 26 - Earthing and Bonding for Electrical Systems.
- C. Do not run HDPE conduit exposed within buildings. Terminations, elbows, stub-ups and extensions of HDPE conduits within buildings shall be made with rigid galvanized heavy-wall steel conduit as specified.
- D. Refer to Section 26 05 33 - Conduit for additional requirements.
- E. Underground telecommunications conduit shall be installed according to Division 27.

2.2 PRECAST CONCRETE MANHOLES AND HANDHOLES

- A. Precast concrete manholes and handholes, of the sizes indicated on Drawings, shall be manufactured under controlled conditions, using steel forms. During manufacture, they

shall not be subject to the elements or to internal or external vibration of concrete. Manholes and handholes shall be subject to periodic testing and control. For concrete requirements, Division 03 Concrete and Section 03 30 00 - Cast-in-Place Concrete.

- B. Steel reinforcing shall be proven by calculations made by a Structural Engineer using new billet steel welded into cages and installed in steel forms before pouring of concrete.
- C. Steel reinforcing shall be as per the manufacturers design 20 kN/m² loading (with calculations provided to the Engineer) using new mild steel welded into cages and installed in steel forms before pouring of concrete.
- D. Concrete class shall be a minimum of C32/40 DC4 and shall be confirmed by the manufacturer.
- E. Install, in each manhole and handhole, four (4) 22 mm diameter formed-steel, hot-dipped galvanized pulling irons. Install inserts for lifting lugs and inserts for channel steel cable racks. Cable rack inserts shall be provided on each wall at intervals not to exceed 450 mm horizontally and 450 mm vertically. Provide cable racks sufficient to support the cables being installed. Hooks shall be all steel or malleable iron and shall be of the removable type. Insulators shall be dry process, glazed porcelain. A preformed hole shall be provided in the base for the future insertion of the earth rod. Ductbank knockout panels shall be provided in each wall equal to a minimum of 50 percent of wall area. If the wall is sectionalized, knockout panels shall be provided in each section equal to 50 percent of section area. Provide a sump pit within chamber that is a minimum 300 x 300 x 150 mm deep to enable a portable sump pump to be used when necessary. Provide corrosion resistant fixed ladders or other suitable climbable devices for each manhole over 1200 mm deep.
- F. Manufacturer shall deliver and Contractor shall set-in-place each precast manhole and handhole. Precast manholes and handholes shall be as manufactured by:
 - 1. Selected and approved EU local Manufacturers.
- G. Manhole and handhole frames and lids shall be cast iron with solid lids, drop handles, and a 900 mm diameter opening. Provide security bolts to prevent unauthorized removal of lid and deliver three (3) wrenches to the Owner's Representative. Bolts to be capped with suitable bolt head caps to provide mechanical protection. Lids shall be identified with cast 100 mm high letters as follows:
 - 1. "ELECTRICAL" for electric manholes and handholes.
- H. Cast iron frames, solid lids and fibrelite covers shall be submitted for approval.
- I. Manhole and handhole joint primer shall be sealed by primary sealant.
- J. Manhole and handhole joint sealer shall be flexible butyl resin sealant.
- K. Precast polymer concrete enclosures, boxes, frames and lids are required to conform with all requirements of EN 1916 and EN 681. Provide matched surface tooling for boxes and covers. Provide covers with a minimum coefficient of friction of 0.05 with identifying Tier level embossed on the top surface. Precast polymer concrete handholes shall not be used in roadways, parking lots or driveways. Precast polymer concrete handholes shall be

manufactured by a selected EU local vendor or approved equal. Thermoplastic polymer and fiberglass/polymer materials are not acceptable.

- L. Where groundwater is within 1000 mm of the formation level of the chamber, a concrete surround shall be provided to the manhole or handhole.

2.3 CAST-IN-PLACE CONCRETE MANHOLES AND HANDHOLES

- A. Cast-in-place concrete manholes and handholes shall be constructed of concrete conforming to Division 3 and sized as indicated on the Drawings. Walls and footings or bases shall be keyed together in a manner to insure watertight construction or shall be placed monolithically where practicable.
- B. Install, in each manhole and handhole, four (4) 22 mm diameter formed-steel, hot-dipped galvanized pulling irons. Install inserts for lifting lugs and inserts for channel steel cable racks. Hooks shall be all steel or malleable iron and shall be of the removable type. A preformed hole shall be provided in the base for the future insertion of the earth rod. Ductbank knockout panels shall be provided in each wall equal to a minimum of 50 percent of wall area. If the wall is sectionalized, knockout panels shall be provided in each section equal to 50 percent of section area. Provide a sump pit within chamber that is a minimum 300 x 300 x 150 mm deep to enable a portable sump pump to be used when necessary.. Provide corrosion resistant fixed ladders or other suitable climbable devices for each manhole over 1200 mm deep.
- C. Conduit, cable racks, pulling-in irons, ladders and all metal shall be secured in position before concrete is placed.
- D. Cast iron frames and lids shall have a Loading class to suit location of covers - vehicle/pedestrian or as similar approved:
 - 1. Selected and approved EU local Manufacturers.
- E. Manhole and handhole frames and lids shall be cast iron or fibrelite with solid lids, drop handles, and minimum 880 mm diameter opening. Provide security bolts to prevent unauthorized removal of lid and three (3) wrenches to the owner's representative. Lids shall be identified with cast minimum 100 mm high letters as follows:
 - 1. "ELECTRICAL" for electric manholes and handholes.
- F. Pulling-in irons shall be steel minimum 22 mm in diameter formed to shape and size with sharp lines and angles.
- G. Cable racks shall be sufficient to accommodate the cables and shall be spaced not more than 450 mm apart horizontally and vertically. The wall bracket shall be channel or section steel. Hooks shall be all steel or malleable iron and shall be of the removable type. Insulators shall be a dry process, glazed porcelain. Cable racks shall comply with the following sections: Section 26 05 36 - Cable Trays for Electrical Systems and Section 26 05 29 - Hangers and Supports for Electrical Systems.
- H. For precast chambers provide a lug on the inside of the chamber tied into the rebar for connection to an earth bar in the chamber.

2.4 EARTH RODS

- A. Refer to Section 26 05 26 - Earthing and Bonding for Electrical Systems. EARTH RODS AND ELECTRODES.

2.5 ARC TAPE

- A. Flame-resistant arc tape for cables shall be 3M Scotch 77 or International Protective Coatings (IPC) Corporation Flamesafe AFT Series for outdoor applications, or approved equal. Tape shall be non-corrosive to cable sheath, shall be self-extinguishing and shall not support combustion. Outdoor tape shall not deteriorate when subjected to oil, water, gasses, salt water, sewage or fungus.

PART 3 EXECUTION

3.1 UNDERGROUND DUCT LINES

- A. Duct lines shall slope downward toward manholes and handholes and away from buildings, with a pitch of not less than 100 mm in 10 m. Top of the concrete envelope shall be not less than 610 mm below finish grade or paving. Conduits shall terminate in end bells where duct lines enter concrete manholes and handholes. Spacers shall be precast concrete or unfilled high-impact polystyrene compound, shall maintain no less than 80 mm separation between ducts and shall be placed on an interval of three (3) spacers per 6000 mm. Provide calculation to determine the required separation between ducts in order to achieve the required ampacity of the cables to be pulled in the duct system. Joints of the conduit shall be staggered by rows and layers between each conduit run so as to provide a duct line having the maximum strength.
- B. During construction, partially completed duct lines shall be protected from the intrusion of debris, such as mud, sand and dirt, by means of suitable conduit plugs. As each section of a duct line is completed, a testing mandrel, not less than 300 mm long with a diameter 60 mm less than the size of the conduit, shall be drawn through each conduit, after which a brush with stiff bristles shall be drawn through until the conduit is clear of all particles of earth, sand or gravel. Conduit plugs shall then be installed immediately. A CCTV camera survey shall also be carried out on completed duct runs at approximately 30 m intervals to ensure there are no cracks in the duct wall which may not be picked up using a mandrel.
- C. Conduit passing under roadways, including paved parking areas, or under or through foundation walls shall be rigid steel or high-density polyethylene (HDPE) conduit, 90 degrees C, 450/750N as per IEC 61386-24 in a reinforced duct bank. Rigid steel conduit or reinforcing shall extend minimum 1500 mm beyond each side of roadway or wall.
- D. Provide conduit end bells at terminations of PVC conduit flush with concrete manhole and handhole walls. Provide insulated earthing bushings when metallic conduit is terminated in a pull box. Provide flexible couplings and expansion joints as recommended by the manufacturer.
- E. Minimum horizontal sweep radius for outside duct run bends shall be minimum 3000 mm unless otherwise indicated within the design package. Conduit stub-ups into site

equipment may be manufactured large radius elbows with bend radius minimum 1200 mm and not less than cable manufacturer's recommendation.

- F. Maintain minimum separation as per local regulations between electric and telecommunications conduit and parallel steam lines underground. Maintain minimum separation as per local regulations between conduits crossing steam lines underground and fill clearance space with foam glass insulation. Maintain minimum separation as per local regulations between telephone/telecommunications conduits crossing electric power conduits underground and fill clearance space with concrete
- G. Provide electric manholes and handholes on maximum 120 m centers in straight conduit runs. The distance between manholes may be increased subject to acceptable results of cable pulling calculations and written approval by the Owner. For each 90-degree bend in conduit run, reduce centers of manholes or handholes by 15 m. A maximum of three (3) 90 degree (or a total of 270 degrees) conduit bends are allowed between vaults and junction boxes.
- H. Seal inside and around all underground conduits entering into buildings at the point of entry into the building against water entry. Do not install any joints or fittings between the point of entry and the conduit water seal. Refer to Section 26 05 44 - Sleeves and Sleeve Seals for Electrical Systems.
- I. For MV chambers, seal inside and around all underground conduits against water ingress at the point of entry into the building or underground chambers containing instrumentation, junction boxes or any other equipment which needs regular maintenance or inspection. Exceptions to the previously stated are sump pump chambers which are designed to remove the water from the ducts.
- J. Where practically possible, all the underground lines shall be installed with a slope towards one of the chambers. If the line is between two chambers then part of the line can have a slope towards one chamber and the second part towards the second chamber. This will allow water removal in case of pulling new cables.

3.2 TRENCH EXCAVATION, BACKFILL AND COMPACTION

- A. Excavation, backfill and compaction shall be in accordance with Division 31.
- B. Backfill shall meet the thermal resistivity as defined by the A/E of Record's calculations and verified by the Contractor.
- C. Where conduit lines are installed under grass plots, the grass shall be removed, stored at one side of the lot and kept watered. As soon as the trench is backfilled, this grass shall be replaced. Where conduit is installed between trees, trees and roots must be protected from damage.
- D. All duct banks of (3) conduits or more shall be covered with (1) sack slurry to prevent air pockets.

3.3 MANHOLES AND HANDHOLES

- A. The contractor shall be responsible for the setting-in-place of each precast concrete manhole and handhole.
- B. Cast-in-place concrete manholes and handholes shall be installed in accordance with Division 03.
- C. All joints shall be primed and sealed.
- D. Apply two (2) coats of heavy-duty coal tar pitch waterproofing compound to all exterior surfaces of concrete manholes and handholes.
- E. Provide manholes sized as shown and in accordance with IEC 60364 (all parts) and handholes in accordance with IEC 60364 (all parts) pull and junction box requirements.
- F. Provide minimum 160 mm stone base for drainage under open bottom polymer concrete handholes. In general, open bottom polymer concrete handholes are allowed only if the bottom sits at least 30 cm above the expected underground water level. In that situation, and where permeability testing allows, provide minimum 160 mm stone/gravel base for drainage under the open bottom. If the bottom would sit below expected underground water level the polymer concrete handholes should be with a sealed bottom and cable entries should be sealed (water tight)

3.4 EARTH RODS

- A. Where required by the contract documents each manhole and handhole shall have an earth rod driven a minimum of 2500 mm into undisturbed earth.
- B. Refer to Section 26 05 26 - Earthing and Bonding for Electrical Systems.

3.5 CABLE INSTALLATION

- A. Pull in cable using approved cable pulling plan, recommended lubricants, and a dynamometer or direct hydraulic potentiometer to monitor cable-pulling tension. Contractor shall also provide recorded pull tensions that verify maximum cable-pulling tension and sidewall bearing pressure for each pull did not exceed the cable manufacturer's written recommendations.
- B. Fireproof medium-voltage cables installed in manholes, handholes, and cable trenches with flame-resistant arc tape extending to 250 mm inside the ducts.
- C. Refer to Section 26 05 13 - Medium Voltage Cable and Section 26 05 19 - Low-Voltage Electrical Power Conductors and Cables for additional details.

END OF SECTION

REVISION HISTORY

Version	Date	Description and location of change
0.1	August 2024	Initial version – BOD Draft submission
0.2	September 2024	BOD Final Submission
0.3	February 2025	Draft IFC Updates